

NAME:

STUDENT ID:

The code below builds a scatter plot using some of the variables in the penguins data frame. Copy this into your session, as you'll be modifying it shortly.

```
ggplot(penguins, aes(x = bill_length_mm,  
                    y = bill_depth_mm,  
                    color = species)) +  
  geom_point() +  
  lims(x = c(30, 60),  
       y = c(12, 23))
```

Don't forget to add a code chunk at the top to load your packages!

Question 1

part a

Extract a data frame from penguins that excludes the Adelie penguins *without using* a data pipeline. Copy the code you used here.

part b

Now, *use a data pipeline* to extract a data frame that excludes the Adelie penguins. Save it into a new object! Copy the code you used here.

part c

Continue the pipeline by piping the data frame you just created to `ggplot()` to create a scatter plot of bill length vs. bill depth. You'll see that you can do data transformations in a pipeline leading up to a single plot, if you don't need to keep the transformed dataframe around once the plot is done.

Question 2

Extract a data frame that excludes the Adelie penguins and retains penguins with bill lengths between 40 and 50 mm. Save the result into a new variable called `adelie_medium`. Copy the code you used here.

Question 3

Sort the data frame from **Question 2a** in decreasing order by bill length. Copy the code you used here.

Question 4

Calculate the mean bill length and bill depth *for each of the three species of penguins*. Copy the code you used here. Do those statistics line up with what you see in the penguins plot provided by the (original) code given?

Question 5

Consider a new metric called `bill_size` that's the sum of the length and depth. What is the average bill size and its standard deviation among each species, broken out among each of the island? Sort your resulting data structure in decreasing order by average bill size. Copy the code you used here.

Question 6

What are the total number of penguins in the data set belonging to each species-island combination?

Question 7

What is the proportion of penguins for each species having a body mass greater than 4000 grams? Copy the code you used here.